AUDIT-SID: Artifact-Level Diagnostics for Semantic-ID Tokenizers in Generative Recommendation

Anonymous Author(s)

Abstract

Semantic-ID tokenizers are a core component of generative recommendation: items are mapped to discrete code sequences and retrieved by sequence generation. Yet most evaluations still emphasize final ranking metrics, which can hide codebook collapse, full-code collisions, collaborative-neighborhood mismatch, tail compression, or expensive prefix structures. We present AUDIT-SID, a mapping-first diagnostic toolkit for item-to-SIDtokenizer artifacts. AUDIT-SIDstandardizes item mappings, metadata, interactions, and optional generator outputs, then reports five main artifact diagnostics: utilization, collision profile, semantic-collaborative alignment, head-tail capacity allocation, and deployment-cost proxies. On public Amazon resource demos, AUDIT-SIDingests a GRID/RQ-KMeans-style export and a ReSID/GAOQ export, with controlled sanity tokenizers for interpretation. A same-item Musical case study on 23,742 items exposes sharply different artifact profiles: a GRID feature-text residual-k-means row has 3,749 unique full SIDs and a 0.9769 full-collision rate, while the bounded ReSID/GAOQ row is collision-free under its exported mapping. The contribution is a reusable artifact audit suite, not a new tokenizer or a full generative-recommender leaderboard.

CCS Concepts

• **Information systems** → **Recommender systems**; *Retrieval models and ranking*.

Keywords

semantic IDs, generative recommendation, tokenizer diagnostics, recommender systems

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1 Introduction

Semantic-ID (SID) tokenization has become a practical design point for generative recommendation. Systems such as TIGER map items into discrete code sequences so that a sequence model can generate candidate identifiers rather than score every catalog item directly [11]. This shift has made the tokenizer itself a first-order artifact. Recent work explores residual quantization frameworks, recommender-native encoding, differentiable SIDlearning, non-uniform quantization, qualified collisions, and continual tokenization [3–5, 7, 9, 14]. The broader identifier literature also shows that item indexing choices matter before generative retrieval is trained end-to-end [6], that semantic IDs can improve industrial ranking generalization [12], and that collaborative tokenization methods explicitly optimize behavioral alignment and assignment diversity [13, 15].

The evaluation practice has not caught up with this artifact boundary. Prior recommender tooling has argued for behavioral tests beyond NDCG-style aggregates [1], but SID tokenizers introduce a distinct artifact: the generated item address space. A tokenizer can support a usable downstream Recall@K while still hiding failure modes that are hard to diagnose from aggregate ranking metrics: underused codebooks, repeated full codes, semantically neat but behaviorally poor prefixes, tail items sharing too little capacity, or prefix structures that increase serving cost. These issues matter because SIDs are not merely features. They define the address space exposed to the generator.

We present AUDIT-SID, a resource for auditing item-to-SID tokenizer artifacts before or alongside downstream recommendation evaluation. The goal is deliberately narrower than proposing a new tokenizer. AUDIT-SIDasks a concrete question: given a public tokenizer artifact that maps items to code sequences, what can be said about its capacity use, collisions, behavioral alignment, head-tail allocation, and deployment cost without retraining a full generator?

The resource contribution is threefold. First, AUDIT-SIDdefines a small mapping-first interface for SIDassignments, item metadata, interaction histories, and optional generator outputs. Second, it implements D1–D5a diagnostics for utilization, collisions, semantic-collaborative alignment, head-tail capacity, and structural serving-cost proxies, with D6 churn support for continual-tokenizer settings. Third, it provides public resource-demo evidence on GRID/RQ-KMeans-style, ReSID/GAOQ, and controlled sanity artifacts, including a same-item Musical case study. The paper follows the CIKM Resource four-page constraint, where appendices count toward the limit and detailed logs are therefore placed in the online artifact [2].

2 Toolkit and Diagnostics

AUDIT-SIDtreats a tokenizer export as an artifact, not as an opaque component of a trained recommender. The minimum input is a table `sid_assignments(item_id, sid_0, ..., sid_L)`. Optional tables provide item metadata, interaction histories, and generator outputs. This contract lets adapters normalize heterogeneous repositories while keeping the audit independent of a particular training loop.

D1: utilization. AUDIT-SIDreports per-level usage, prefix counts, entropy-style imbalance summaries, and dead or underused code

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Tokenizer exports: repo SIDs, public mappings, or controlled sanity tokenizers

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Artifact contract: `sid_assignments`, `item_metadata`, `interactions`, optional `generator_outputs`

↓

Validation: join coverage, prefix depth, provenance, and adapter caveats

↓

Diagnostics: D1 utilization; D2 collisions; D3 collaborative alignment; D4 head-tail; D5a structural cost; optional D6 churn; future D7 generator behavior

↓

Outputs: CSV/Markdown/LaTeX audit tables and failure-case slices

Figure 1: AUDIT-SIDartifact pipeline. The current resource implements D1–D5a over item-to-SIDmappings, includes optional D6 churn support, and reserves D7 for artifacts that expose generated candidates or beam traces.

indicators. These expose codebook collapse and uneven capacity even when full ranking metrics are unavailable.

D2: collision profile. AUDIT-SIDmeasures duplicate full SIDs and prefix collisions at each depth. In the current resource version, D2 is a collision profile rather than a causal harm estimate. Interaction-qualified collision harm, as motivated by qualification-aware collision work [5], is reserved for the next artifact version.

D3: semantic-collaborative alignment. Semantic grouping and collaborative usefulness need not agree. AUDIT-SIDtherefore computes a co-occurrence-based collaborative neighborhood proxy and asks how often prefix neighborhoods recover those behaviorally related items. Metadata purity is retained only as an auxiliary diagnostic, not as the definition of collaborative quality.

D4: head-tail capacity. AUDIT-SIDsplits items into popularity buckets and measures whether full SIDcapacity and prefix structure are allocated differently across head, mid, and tail items. This helps separate genuine tokenizer behavior from simply restating the catalog popularity distribution.

D5a, D6, and D7. D5a reports structural cost proxies: SIDlength, prefix fan-out, duplicate codes, and trie-like expansion pressure. These are not real generator serving latencies. D6 computes churn between tokenizer refreshes and is useful for drift-aware continual-tokenization settings [3]. D7 is an interface hook for generator behavior, such as invalid paths, next-token entropy, and duplicate generated candidates; it requires `generator_outputs` and is not claimed by the current evidence.

Identifier foundations and recommender diagnostic frameworks motivate the resource framing, but they are not runnable tokenizer rows. Table 1 therefore focuses on SID artifact facets and marks which parts are demonstrated in v0.

3 Resource Demo and Case Study

The demo is designed as a diagnostic-resource check rather than a leaderboard. It asks whether AUDIT-SIDcan ingest real public SIDartifacts from both a canonical residual-quantization line and a recent recommender-native line, then produce comparable audit tables with explicit caveats. GRID provides the open RQ-KMeans-style path [7]; ReSID provides the GAOQ line and processed Musical artifacts [9]. Sanity tokenizers are kept as controlled references for metric sensitivity.

Table 2 compares two exports on the same 23,742 Musical items. The GRID row is intentionally labeled as feature-text: it uses processed feature text from the ReSID artifact path and the official MiniBatchKMeans-style residual quantization module, not a faithful raw-text TIGER reproduction. Under that controlled setup, the row exposes high collision pressure and limited tail capacity. The ReSID/GAOQ row exports one full code per item in this bounded run, yielding zero full-code collisions and higher D3-L1 collaborative prefix recovery.

This contrast is useful precisely because the table is not a downstream ranking claim. It shows that the same audit interface can surface different artifact mechanisms: compact residual-k-means codes can collapse many items into shared full SIDs; recommender-native GAOQ can avoid full collisions in the exported mapping; and the head-tail table makes the capacity allocation visible rather than inferred from item popularity. Additional artifact tables cover sanity controls, GRID scale checks, DACT churn, and a MovieLens schema smoke, but those are placed in the repository rather than the four-page PDF.

4 Availability and Limitations

The artifact repository contains the AUDIT-SIDPython package, adapter scripts, metric runners, generated CSV, Markdown, and LaTeX tables, provenance logs, and the BibTeX/source audit used for this draft. Large local artifacts are kept out of git and described through manifests so that reviewers can distinguish public reproducibility assets from local caches.

The current evidence supports a conservative resource-demo claim. It does not support several stronger claims. AUDIT-SIDdoes not reproduce TIGER end-to-end; the GRID Musical row is a controlled feature-text export. The ReSID Sports balanced GAOQ path was stopped because constrained k-means was CPU-bound under the available window, so the main ReSID evidence is the bounded Musical export. CARD is repaired only as a fallback/stressor path and should not be presented as faithful CARD reproduction [14]. D2 is a collision profile, not strict causal harm. D3 is a diagnostic proxy and is not yet proven monotonic with Recall@K or NDCG. D5a is a structural cost proxy without real generator-output latency, and D6 remains optional drift evidence.

These boundaries are intentional. The paper contributes an audit surface for tokenizer artifacts, not a new SIDtokenizer, a complete generative-recommender evaluation, or an industrial online-impact study. Future versions should add interaction-qualified collision harm, generator output diagnostics, stronger same-dataset method coverage, and unified search and recommendation artifact checks [10]. Industrial SID deployment studies further motivate tracking these artifact properties as first-class engineering objects [8].

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Table 1: Method/facet coverage in the current AUDIT-SIDresource. R0 foundation and diagnostic-tooling citations motivate the problem but are not listed as runnable tokenizer artifacts.

Facet	Anchor examples	v0 artifact role	Diag.	Boundary
A canonical SID	GRID/RQ-KMeans	runnable main A	D1–D5a	Musical row uses processed feature text, not raw-text TIGER.
B1 collaborative / predictability	ReSID/GAOQ	runnable main B	D1–D5a,D3	Bounded Musical export; not representative of all B facets.
B2 collision / capacity	QuaSID, AdaSID, CARD, CapsID	literature/backlog	D1,D2,D4	Motivates diagnostics; no faithful named-method result claimed.
B3 ranking / retrieval	DIGER, joint search-rec SID	interface target	D3,D7	D7 needs generated candidates or beam traces.
B4 bottleneck / interface	AsymRec, CapsID, Long SID	literature target	D4,D5a,D7	Current v0 has structural D5a only.
C drift / staleness	DACT, SID staleness	optional smoke	D6	Drift evidence, not a Cluster B replacement.
D deployment / search	Snapchat SID, search-rec SID	motivation target	D5a,D7	No production serving or online-impact claim.
Controls / portability	sanity, MovieLens smoke	runnable controls	D1–D5a	Metric checks only; not SID methods.

Table 2: Same-item Musical artifact profiles. This is not a downstream leaderboard; GRID feature-text uses ReSID processed feature text rather than a faithful raw-text TIGER reproduction.

System	Unique	Full coll.	D3 L1	Tail cap.
GRID feature-text	3,749	0.9769	0.0552	0.3695
ReSID GAOQ	23,742	0.0000	0.1535	1.0000

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GenAI Usage Disclosure

The authors used generative AI tools for drafting assistance, code-review support, and experiment-log organization. All claims, tables, and citations in the submission should be verified against the public artifact repository and primary sources before final submission.